Andreas W. Neitzel

PhD candidate in astronomy. Machine learning for Galactic archaeology.

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Education

PhD **Faculdade de Ciências da Universidade do Porto and IA/CAUP**, Astronomy — Porto, Portugal, Sept 2022 – present
- Thesis: Machine Learning Applied to Galactic Archaeology
- Advisors: Tiago L. Campante, Diego Bossini, Andrea Miglio

MSc **Faculdade de Ciências da Universidade do Porto**, Astronomy and Astrophysics — Porto, Portugal, Sept 2020 – Dec 2022
- Thesis: Determination of Age and Mass for Seismic Stars in the Ariel Input Catalog
- Final grade 16/20

BSc **Faculdade de Ciências da Universidade do Porto**, Physics — Porto, Portugal, 2017 to 2020

Research positions

INAF-OAS Bologna, Visiting PhD Student, INAF Osservatorio di Astrofisica e Scienza dello Spazio — Bologna, Italy, Sept 2025 – Mar 2026
Six-month secondment in A. Miglio's group building simulation pipelines for the proposed HAYDN mission.

IA/CAUP, PhD Researcher, Stellar Astrophysics Group — Porto, Portugal, Sept 2022 – present
Doctoral research on machine-learning methods for Galactic archaeology with Gaia, APOGEE and asteroseismic data.

Publications

Dissecting stellar populations with manifold learning. I. Validation of the method on a synthetic Milky Way-like galaxy — Mar 2025
A. W. Neitzel, T. L. Campante, D. Bossini, A. Miglio
10.1051/0004-6361/202451718 (Astronomy and Astrophysics, 695, A243)

Expanding the frontiers of cool-dwarf asteroseismology with ESPRESSO: detection of solar-like oscillations in the K5 dwarf epsilon Indi — Mar 2024
T. L. Campante, H. Kjeldsen, Y. Li, M. N. Lund, *et al. including A. W. Neitzel*
10.1051/0004-6361/202449197 (Astronomy and Astrophysics, 683, L16)

Ariel stellar characterisation. I. Homogeneous stellar parameters of 187 FGK planet host stars: description and validation of the method — July 2022
L. Magrini, C. Danielski, D. Bossini, *et al. including A. W. Neitzel*
10.1051/0004-6361/202243405 (Astronomy and Astrophysics, 663, A161)

Talks and posters

- Oral: JANUS: a neural network for APOGEE-like chemistry of the TESS asteroseismic red giants from Gaia XP spectra, with per-star reliability. IMAS13, 13th Iberian Meeting on Asteroseismology, Cádiar (Granada), Spain, 2026-10.
- Poster: HAYDNsim: Forecasting photometric noise in the crowded fields that defeat current missions. TASC10/KASC17, 10th TESS and 17th Kepler Asteroseismic Science Consortium Workshop, Aarhus, Denmark, 2026-07.
- Oral: Contrastive deep learning for structure-preserving chemistry of TESS asteroseismic red giants from Gaia XP spectra. TASC10/KASC17 workshop, Aarhus, Denmark, 2026-07.
- Oral: Identification of stellar populations through latent space projection. TASC8/KASC15, 8th TESS and 15th Kepler Asteroseismic Science Consortium Workshop, Porto, Portugal, 2024-07.

Grants, fellowships and computing allocations

PhD research fellowship under the FCT project ArqueoGal (CIAAUP-17/2026-BI-D) Sept 2026 – Aug 2027
Fellow. FCT, through project 2024.15303.PEX, hosted at IA

FCT Exploratory Research Project ArqueoGal: Unraveling the evolutionary history of the Milky Way with Galactic Archaeology (2024.15303.PEX) Feb 2026 – Aug 2027
Participant. Fundação para a Ciência e a Tecnologia. Named participant and developer of the project's machine-learning pipelines, with allocations on the Stellar Population Classification and Galactic Modeling tasks.

FCT Doctoral Fellowship (2025.01843.BD) Sept 2025 – Aug 2026
Fellow. Fundação para a Ciência e a Tecnologia

Awards

Best Master's Student in Astronomy and Astrophysics Faculdade de Ciências da Universidade do Porto 2023

Software

Starfold v0.2.0, MIT, sole developer Apr 2026 – present
Python package that finds groups in a table of measurements without labels, using UMAP, HDBSCAN and Optuna, and reports how far to trust each group. It extends Neitzel et al. 2025.

Teaching, supervision and mentoring

- Supervisor, IA Summer Programme 2025. IA/CAUP, Porto, Portugal, 2025-06.

Skills

Python engineering for research software: Typed Python packages with tests, continuous integration, locked environments, provenance stamps, Sphinx documentation.

Probabilistic deep learning: conditional normalising flows, ensembles, mixture uncertainties: An ensemble of flow-based neural networks that returns, for each star, a joint distribution over 25 stellar labels and how much the networks disagree.

Uncertainty calibration and reliability: isotonic calibration, gradient boosting, selection tests, coverage: A calibrated per-star probability that a prediction is wrong (cross-fitted gradient boosting with isotonic calibration), checked with interval-coverage and selection tests.

Out-of-distribution detection: nearest-neighbour novelty in a Gaussianised input space: A per-star novelty score from the distance to the training set, tested on survey samples outside the training selection.

Data engineering at scale: archive ingestion, Parquet pipelines, a manifest-driven data store with provenance: Ingestion of whole Gaia releases into a columnar data store, with manifest-resolved datasets, duplicate checks and provenance on every product.

Unsupervised learning and manifold methods: UMAP, HDBSCAN, Optuna search, stability and recurrence measurement, matched-null calibration.

High-performance computing: Slurm jobs on the ARM, x86 and GPU partitions of Deucalion and on the University of Bologna clusters, for neural-network training, catalogue-scale scoring and photometric-noise simulations. CUDA and RAPIDS on GPUs.

Survey data and stellar modelling tools: Gaia DR3 (astrometry, RVS, XP spectra, GSP-Spec), APOGEE, LAMOST, GALAH, 2MASS, and TESS and Kepler/K2 asteroseismology. Tools: astropy, galpy, TOPCAT, ADQL, PARAM, MESA, TRILEGAL, MWDUST.